Rohit Neelakanta

Bengaluru, Karnataka | rohitrb2405@gmail.com | +91-6360799715

EDUCATION

NMIT (Nitte Meenakshi Institute of Technology) — Bengaluru, Karnataka
B.E. in Electronics and Communication Engineering (ECE) | CGPA: 8.13 — *2023 – 2027*

- **Relevant Coursework:** Engineering Mathematics,C/C++, Digital System Design, Machine Learning

TECHNICAL SKILLS

- **Programming & Scripting:** Python (Pandas, NumPy, Matplotlib, scikit-learn), C, C++
- **Data Engineering & Databases:** PostgreSQL, Advanced SQL (CTEs, Window Functions), SQLAlchemy, Docker, FastAPI
- **Data Analysis & Visualization:** Power BI, Tableau, Advanced Excel
- **Engineering Tools:** Xilinx Vivado, Odoo

DATA ANALYTICS PORTFOLIO PROJECTS

Financial Data Pipeline & Anomaly Detection — *Python, PostgreSQL, Docker, scikit-learn*

- Engineered an automated financial data pipeline utilizing Python and FastAPI to ingest, process, and structure real-time market data via the yfinance API.
- Designed a robust PostgreSQL database schema using SQLAlchemy to systematically track assets, historical price records, and market irregularities.
- Integrated scikit-learn models to automate anomaly detection on time-series data, containerizing the entire application ecosystem with Docker for scalable deployment.

End-to-End E-Commerce Sales Dashboard — *Python, SQL, Power BI*

- Extracted and sanitized a raw dataset of 10,000+ e-commerce transactions using Python and Pandas to eliminate duplicates and handle missing parameters.
- Designed a relational SQL database to store cleaned data, utilizing Common Table Expressions (CTEs) to segment information by region and product category.
- Built an interactive Power BI dashboard to visualize revenue trends, isolating a key drop in seasonal sales to present actionable recommendations for Q4 performance.

Healthcare Digital Twin Data Synthesis — *Python, Data Extraction, Synthesis*

- Systematically extracted and processed qualitative and quantitative data from over 30 academic research papers focusing on personalized healthcare simulations.
- Analyzed and categorized key performance metrics of various AI models, including multiscale CNNs achieving 99.2% accuracy for ECG classification and SVMs achieving 98.7% for Parkinson's diagnosis.
- Synthesized unstructured research findings into structured comparative tables, highlighting statistical trends in data integration, real-time monitoring, and predictive analytics.

ENGINEERING & DEVELOPMENT EXPERIENCE

Major Project Intern – URSC, ISRO — *March 2026 – Present*

- Engineered a flexible and scalable FPGA-based serial communication interface for the testing and validation of On-Board Computer Modules.
- Analyzed complex hardware data logs, identified system anomalies, and verified data integrity across multiple testing phases to ensure accurate module validation.

Backend Data Developer – EcoFinds (Odoo Hackathon) — *September 2025*

- Collaborated on a cross-functional team to develop EcoFinds, a sustainable second-hand marketplace prototype application.
- Structured the backend database using SQL and Python to process user inputs, manage inventory, and ensure seamless data retrieval.